

GARDEN CITY, KS, USA

WMO: 724515

Lat: 37.927N

Lon: 100.725W

Elev: 2882

StdP: 13.23

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 23064

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	3.6	8.9	-4.2	4.9	8.9	0.4	6.2	15.3	34.6	41.5	31.1	39.7	11.1	10	0.675

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	25.9	99.4	69.1	96.9	69.0	93.6	68.8	73.6	89.4	72.4	88.3	71.3	87.2	15.7	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
69.8	122.0	79.5	68.1	114.9	78.0	66.5	108.8	76.9	39.1	89.2	38.0	88.5	37.0	87.3	79.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
32.0	27.8	24.9	-4.9	103.6	5.2	2.7	-8.7	105.5	-11.7	107.1	-14.7	108.6	-18.5	110.6		
			-5.6	75.9	5.1	1.6	-9.2	77.0	-12.2	78.0	-15.0	78.8	-18.7	80.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	55.0	31.9	35.2	44.3	53.1	63.7	74.1	78.7	76.8	69.2	55.9	42.8	32.6
	DBStd	19.04	10.31	11.64	10.95	9.65	8.73	6.85	5.25	5.42	8.02	9.33	9.59	9.97
	HDD50	2148	562	425	237	77	6	0	0	0	2	46	253	540
	HDD65	5035	1025	834	643	369	131	10	1	1	49	304	665	1003
	CDD50	3966	1	11	59	170	431	724	890	830	579	231	38	2
	CDD65	1380	0	0	1	12	91	285	426	366	176	23	0	0
	CDH74	16680	0	13	92	352	1294	3386	5078	3960	2039	440	26	0
	CDH80	8393	0	2	16	111	547	1729	2820	2072	943	151	2	0
Wind	WSAvg	12.3	11.4	12.4	13.4	14.5	13.3	12.9	11.3	10.8	12.1	12.4	12.0	11.4
Precipitation	PrecAvg	19.7	0.4	0.6	1.5	1.7	3.1	3.3	3.0	2.4	1.5	0.9	0.8	0.5
	PrecMax	30.3	1.3	1.9	7.2	4.8	8.0	9.4	10.4	6.9	4.2	4.2	3.7	1.6
	PrecMin	10.3	0.0	0.0	0.0	0.0	0.2	0.1	0.4	0.1	0.1	0.0	0.0	0.0
	PrecStd	4.8	0.4	0.6	1.6	1.3	1.7	2.3	2.2	1.6	1.3	1.0	0.9	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	68.3	75.0	81.8	89.3	97.1	102.7	102.3	102.0	98.4	90.8	77.5	66.3
		MCWB	46.9	50.4	55.1	57.5	63.2	66.3	69.0	68.9	66.4	61.3	53.0	48.9
	2%	DB	61.4	67.4	75.9	82.9	91.0	97.4	99.6	97.7	93.7	83.9	72.0	60.0
		MCWB	44.4	47.3	52.8	57.1	63.3	68.0	70.1	70.0	66.6	60.0	51.8	44.5
	5%	DB	54.9	60.8	69.9	77.8	86.0	93.5	97.1	94.5	90.0	78.9	65.8	54.3
		MCWB	41.3	44.7	50.9	55.3	62.4	68.0	70.1	69.9	66.2	58.4	49.8	41.9
	10%	DB	48.5	54.1	63.8	72.1	81.5	90.2	93.6	91.1	85.6	73.0	60.0	48.8
		MCWB	38.2	41.6	48.9	54.0	61.9	68.2	70.4	69.6	65.5	56.8	46.5	39.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.5	52.5	60.2	62.7	69.5	74.1	75.4	75.1	71.8	67.0	57.4	50.6
		MCDB	64.7	68.1	71.6	75.2	83.8	88.5	90.9	89.2	87.1	78.9	68.3	62.4
	2%	WB	44.8	48.8	56.2	60.0	67.3	72.3	74.1	73.2	70.0	64.1	53.9	45.8
		MCDB	59.9	65.5	67.8	74.3	81.0	87.5	90.1	89.0	85.4	74.8	64.7	58.4
	5%	WB	41.9	45.7	52.9	57.8	65.7	70.9	72.8	72.0	68.5	61.4	51.2	42.3
		MCDB	54.1	59.5	66.1	72.3	79.3	86.3	89.0	87.9	83.4	71.9	64.3	53.2
	10%	WB	38.9	42.2	49.8	55.4	63.8	69.5	71.7	70.9	67.1	58.8	48.1	39.4
		MCDB	48.0	53.6	63.7	69.3	77.4	84.3	87.7	86.3	81.1	70.8	59.9	48.1
Mean Daily Temperature Range	5% DB	MDBR	25.5	26.1	27.9	27.5	26.5	26.3	25.9	24.8	26.3	27.3	27.1	24.9
		MCDBR	36.7	38.0	38.1	37.6	33.8	31.5	30.6	30.2	31.8	35.4	36.5	33.8
		MCWBR	21.9	21.1	19.1	16.5	12.8	10.1	9.0	9.4	10.6	15.3	19.4	20.8
	5% WB	MCDBR	34.3	35.5	33.0	30.7	28.2	27.2	26.6	26.2	26.6	26.9	32.4	32.0
		MCWBR	21.1	20.4	18.1	15.4	12.8	10.9	9.7	9.6	9.9	13.6	18.0	20.2
Clear-Sky Solar Irradiance	taub	0.260	0.268	0.299	0.328	0.348	0.372	0.390	0.372	0.342	0.302	0.271	0.259	
	taud	2.551	2.554	2.469	2.407	2.422	2.392	2.376	2.385	2.454	2.544	2.565	2.559	
	Ebn at Noon	300	310	307	302	295	287	281	284	288	292	293	292	
	Edh at Noon	23	27	32	36	36	38	38	37	32	27	23	22	
All-Sky Solar Radiation	RadAvg	844	1104	1503	1844	2137	2339	2298	2008	1687	1242	928	745	
	RadStd	73	112	130	103	130	120	87	107	97	137	80	55	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (17 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page